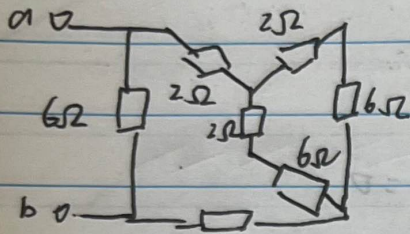
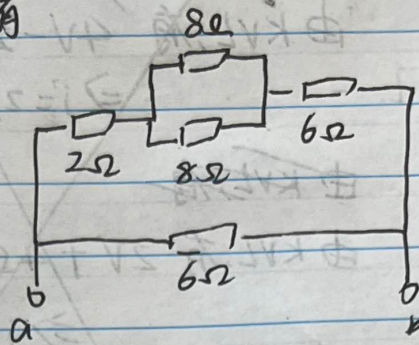


第二次作业

2.2 Δ 型电阻转换为Y型电阻有



即为



$$8\Omega // 8\Omega = 4\Omega$$

$$6\Omega // (2 + 4 + 6)\Omega = 4\Omega$$

$$R_{eq} = 4\Omega$$

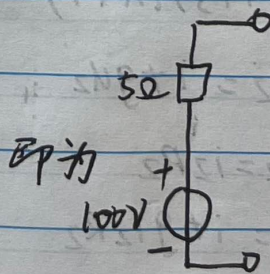
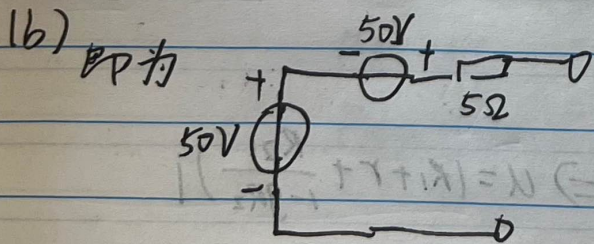
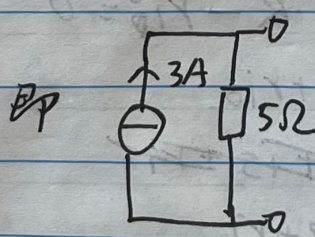
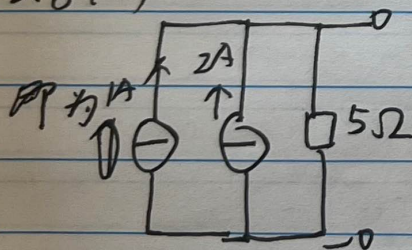
2.4 R_{eq}

$$\therefore \frac{U_2}{U_1} = 0.05 \quad \therefore \frac{R_2 // U_2}{R_1} = \frac{1}{19}$$

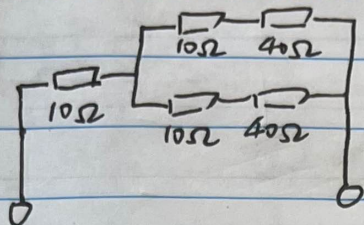
$$\therefore R_{eq} = 40\text{K}\Omega \quad \therefore R_1 = 38\text{K}\Omega \quad R_2 // U_2 = 2\text{K}\Omega$$

$$\text{即 } \frac{R_2 \cdot 5\text{K}\Omega}{R_2 + 5\text{K}\Omega} = 2\text{K}\Omega \Rightarrow R_2 = \frac{10}{3}\text{K}\Omega$$

2.8 (a)



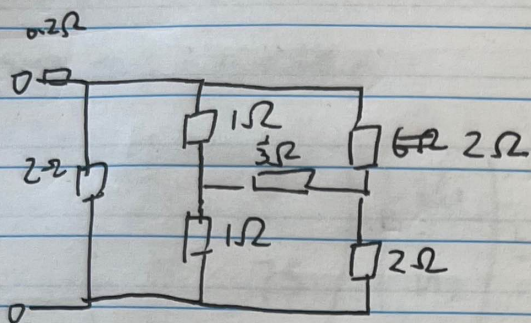
2.9 $\Delta \rightarrow Y$ 型, 即



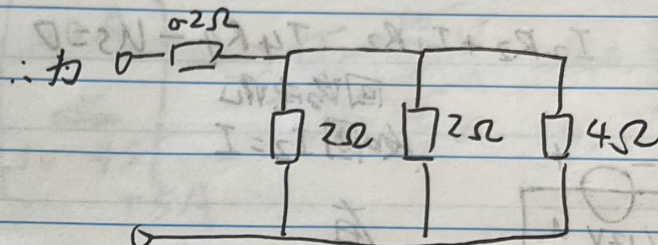
$$R_{eq} = 10\Omega + \frac{1}{\frac{1}{50} + \frac{1}{50}} \Omega = 35\Omega \quad (1)$$

1b) 电阻力

将所有的
并联电阻
等效有

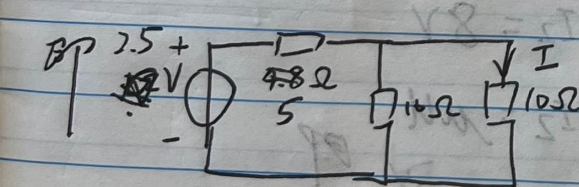
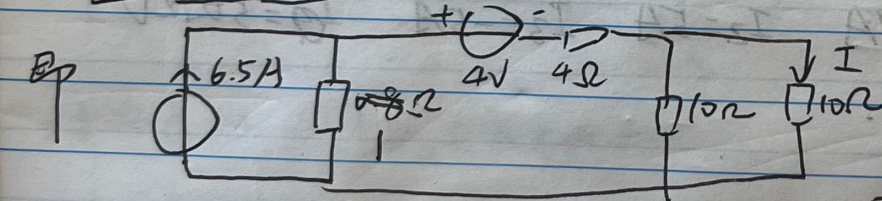
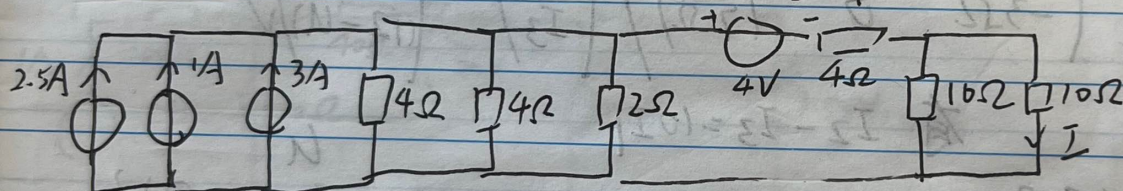


显然图中电桥无效



$$R_{eq} = 0.2\Omega + \frac{1}{\frac{1}{2} + \frac{1}{2} + \frac{1}{4}} \Omega = 1.2\Omega$$

2.10 电压源换电流源, 电流源换电压源



$$I = \frac{2.5V}{\frac{8\Omega \times 10\Omega}{8\Omega + 10\Omega} + 10\Omega} = 0.125A$$

2.11 (b) 对独立节点列 KCL

$$①: I_5 + I_2 - I_1 - I_3 = 0$$

$$②: I_3 - I_2 - I_4 = 0$$

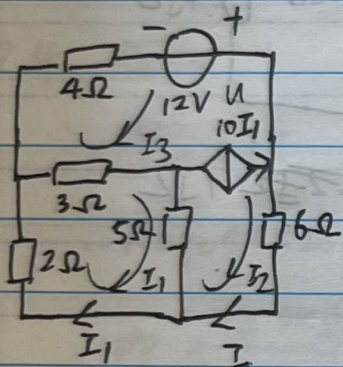
$$③: I_1 + I_4 - I_5 = 0$$

对网孔列 KVL, 有

$$I_2 R_2 + I_3 R_3 - U_5 = 0$$

$$I_2 R_2 + I_1 R_2 - I_4 R_4 - U_5 = 0$$

2.14



回路电流
如图 $I_2 = I$

有

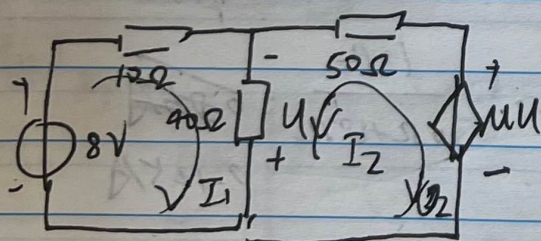
$$\begin{pmatrix} (2+3+5)\Omega & -5\Omega & -3\Omega \\ -5\Omega & (5+6)\Omega & 0 \\ -3\Omega & 0 & 17\Omega \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{pmatrix} 0 \\ U \\ (12-U)V \end{pmatrix}$$

$$\text{有 } I_2 - I_3 = 10I_1$$

$$\text{联立解得 } I_1 = 1A \quad I_2 = 5A \quad I_3 = -5A \quad U = 50V$$

$$\therefore I = I_2 = 5A$$

2.16.



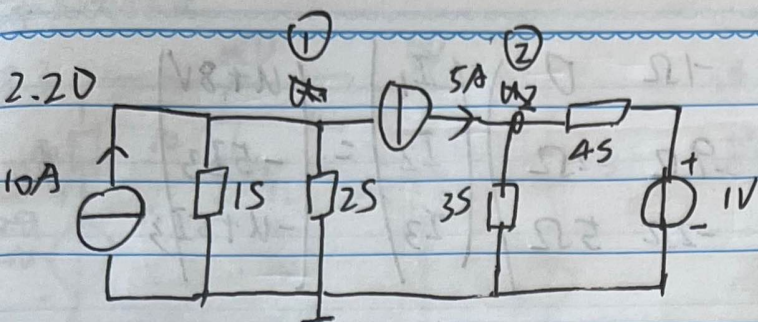
$$\text{有 } 50\Omega \cdot I_1 + 40\Omega \cdot I_2 = 8V$$

$$40\Omega \cdot I_1 + 90\Omega \cdot I_2 = U$$

$$U = (I_1 + I_2) \cdot 40$$

$$\text{即 } \begin{pmatrix} 50 & 40 & 0 \\ 40 & 90 & -40 \\ 40 & 40 & -1 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ U \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}$$

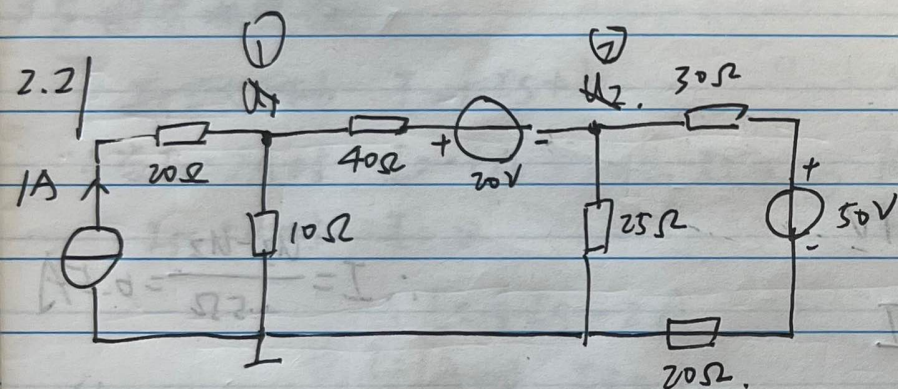
$$\text{当 } \det \begin{pmatrix} 50 & 40 \\ 40 & 90 \end{pmatrix} \neq 0 \text{ 时}$$



对节点①②, 节点电压方程为

$$\begin{pmatrix} 3S & 0 \\ 0 & 7S \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = \begin{pmatrix} 10A \\ -5A \end{pmatrix} \Rightarrow \begin{pmatrix} 10 \\ -5 \end{pmatrix} \Rightarrow \begin{pmatrix} U_1 = \frac{10}{3} V \\ U_2 = \frac{-5}{7} V \end{pmatrix}$$

$$\therefore P = (U_2 - U_1) \cdot 5A = -1.9W$$



对节点①, ② 节点电压方程为

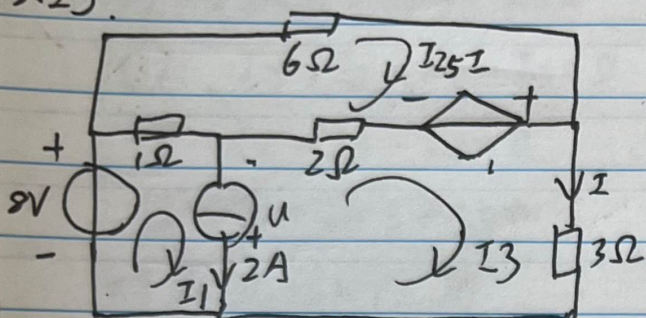
$$\begin{pmatrix} \frac{1}{20} + \frac{1}{10} + \frac{1}{40} & -\frac{1}{40} \\ -\frac{1}{40} & \frac{1}{40} + \frac{1}{25} + \frac{1}{20} + \frac{1}{50} \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = \begin{pmatrix} 1 + \frac{20}{40} \\ \frac{50}{20} - \frac{20}{40} \end{pmatrix} \Rightarrow \begin{pmatrix} U_1 = 14V \\ U_2 = 10V \end{pmatrix}$$

$$\therefore U_S = U_1 + IR = 34V$$

$$P = U_S I = 34W$$

求解, 即 $U = \frac{29}{2}$

2.23



$$\text{有} \begin{pmatrix} 1\Omega & -1\Omega & 0 \\ -1\Omega & -9\Omega & -2\Omega \\ 0 & -2\Omega & 5\Omega \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{pmatrix} U+8V \\ -5I_3 \\ -U+5I_3 \end{pmatrix}$$

$$I_1 - I_3 = 2A$$

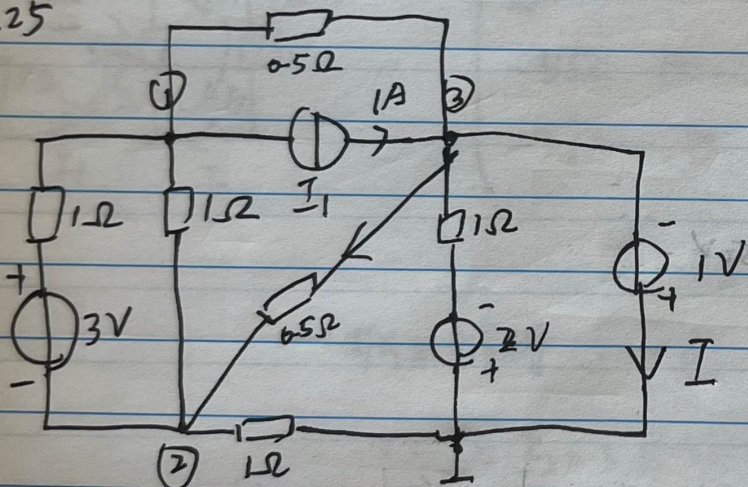
$$\text{联立有 } I_1 = 6A \quad I_2 = -\frac{2}{3}A$$

$$I_3 = 4A \quad U = -\frac{4}{3}V$$

$$\therefore P_u = u_s I_1 = 48W$$

$$P_i = U \cdot I = -\frac{8}{3}W$$

2.25



$$\therefore I = \frac{U_3 - U_2}{0.5\Omega} = 6.5$$

①节点有

$$\begin{pmatrix} \frac{1}{1} + \frac{1}{1} + \frac{1}{0.5} & -1 & -1 \\ -1 & \frac{1}{1} + \frac{1}{1} + \frac{1}{0.5} + \frac{1}{1} & -\frac{1}{0.5} \\ -2 & -\frac{1}{0.5} & \frac{1}{0.5} + \frac{1}{0.5} + \frac{1}{1} \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \\ U_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \\ 1 - \frac{2}{1} - 1 \end{pmatrix}$$

$$\text{有 } U_3 = -1V$$

$$\Rightarrow \cancel{U_1 = 0} \quad U_1 = -\frac{5}{8}V \quad U_2 = -\frac{5}{4}V$$